Causally Robust Reward Learning with Reason-Augmented Preference Feedback

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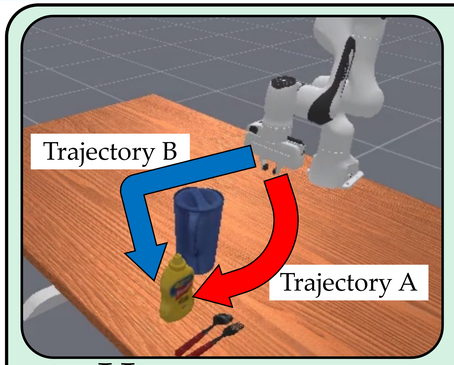
Motivation

- **Preference-based reinforcement learning** shapes agent behaviors from user's binary preferences, which can leave it vulnerable to **causal confusion**.
- The learned reward can latch onto **spurious features** that cooccur with preferred trajectories, collapsing when those correlations disappear at test time.

Insight: A natural language reason can (1) clarify true causal signals behind preference and (2) improve generalization beyond spurious features.



Could you tell me which trajectory you prefer?

I prefer Trajectory A, **because it smoothly avoids the obstacle**.

How can we use this **reasoning** information?

***TL;DR* With reasons behind preference, we can improve robustness of reward models and downstream policies!**

ReCouPLe: Reason-based Confusion Mitigation in Preference Learning

Preliminary:

- We learn a trajectory encoder ϕ that maps a trajectory τ into the embedding space of the pretrained language encoder (T5 sentence embedding).
- We represent each task's reward given its task description ℓ_{task} as **an inner product** of $\phi(\tau)$ and task embeddings $\theta = \text{LM}(\ell_{\text{task}})$:

$$r(\tau, \ell_{\text{task}}) = \phi(\tau)^\top \theta$$

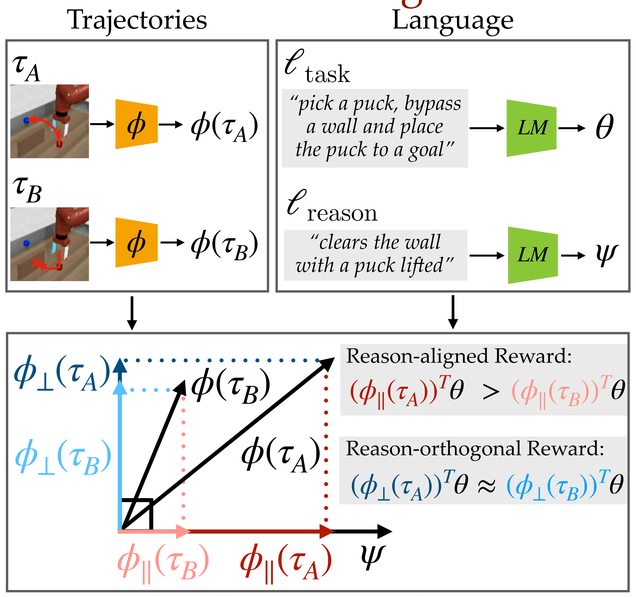
ReCouPLe treats the rationale embedding ψ as **a projection axis**, splitting the trajectory representation into reason-aligned and reason-orthogonal parts:

$$\phi_{\parallel}(\tau) = \left(\frac{\phi(\tau)^\top \psi}{\|\psi\|_2^2}\right)\psi, \quad \phi_{\perp}(\tau) = \phi(\tau) - \phi_{\parallel}(\tau)$$

Then, we can decompose the reward into two components:

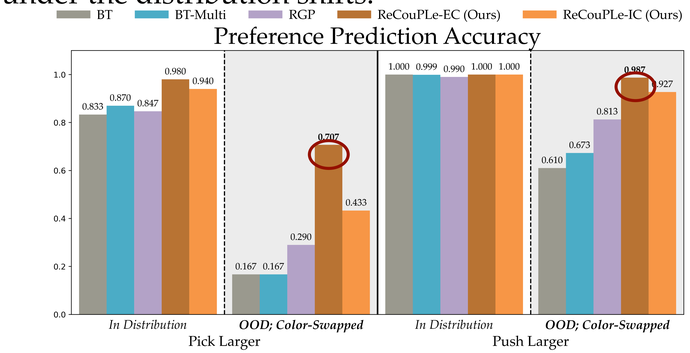
$$r(\tau, \ell_{\text{task}}) = \phi_{\parallel}(\tau)^\top \theta + \phi_{\perp}(\tau)^\top \theta$$
$$= \underbrace{r_{\parallel}(\tau)}_{\text{reason-aligned}} + \underbrace{r_{\perp}(\tau)}_{\text{reason-orthogonal}}$$

Should mainly determine preference | Should not affect preference as much

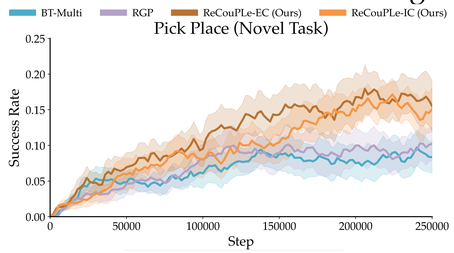


Results

1. **ReCouPLe** robustly predicts user's preference under the distribution shifts.



2. **ReCouPLe** transfers to a novel task, thanks to shared semantics between training and unseen tasks.



Reward: trained with preference data of three manipulation tasks in Meta-world benchmark.

Policy: trained on the offline play data of a semantically similar, novel task (result shown left).

3. Additional experiments show that **ReCouPLe** is (i) scalable to sparse reason settings and (ii) robust to linguistic diversity in reasons & different dataset sizes.